

Hrithik Sham D H

hrithikksham@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | +91 9500677722

SUMMARY

Backend-heavy full stack engineer (fresher) who builds and ships — **RAG pipelines**, **vector search systems**, and **APIs handling 500+ concurrent requests** in production. Strong bias toward clean architecture and measurable performance and security over code that just works.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, SQL

Frameworks: FastAPI, Node.js (Express), React.js, React Native (Expo), LangChain

Databases: PostgreSQL, MongoDB, Redis, Qdrant, Supabase

Tools & Cloud: Docker, AWS, Nginx, Git/GitHub, Linux, Sentence Transformers

Core Concepts: System Design, REST APIs, JWT Auth, RBAC, RAG, Vector Search, Caching, Microservices

PROJECTS

Adaptive AI Memory Engine | *FastAPI, Qdrant, Supabase, Sentence Transformers, Groq* Jan 2026 – Present

- Designed a **novel 4-state memory lifecycle** (FRESH → ACTIVE → FADING → ARCHIVED) inspired by Ebbinghaus forgetting-curve theory — archived memories auto-excluded from retrieval, keeping results relevant without manual curation.
- Built a **live semantic retrieval pipeline** (encode → cosine search → threshold filter → ranked top-K) using all-MiniLM-L6-v2 with Groq-generated contextual answers; **clean 5-layer service architecture**, each module independently replaceable.

Autonomous LLM Agent | *React, FastAPI, MongoDB, LangChain, RAG* July 2025 – Present

- Reduced API adoption effort by **~40%** by building a **RAG-powered assistant** that interprets live documentation and generates grounded, executable code — hallucination-resistant by design.
- Sustained **sub-2s latency** under high-concurrency load by layering response caching into the reasoning pipeline, achieving production-grade stability without scaling costs.

URL Shortener & Analytics Platform | *Node.js, PostgreSQL, Redis, Prisma* Oct 2025 – Dec 2025

- Served **~70% of all requests** from Redis cache with **sub-50ms** average response time; hardened against traffic bursts via **token-bucket rate limiting** — zero degradation under abusive load.

Yournote App | *React Native, FastAPI, MongoDB, Redis* Jan 2025 – June 2025

- Shipped a cross-platform mobile app with **10,000+ online-synced notes**, JWT auth, and encrypted local storage; cut API call volume by **~35%** via Redis caching and MongoDB schema redesign.

WORK EXPERIENCE

Backend Developer Intern Aug 2025 – Oct 2025

InnoByte Services

- Drove a **40% throughput gain** under **500+ concurrent requests** by combining DB indexing, cursor-based pagination, and Nginx reverse-proxy load balancing — no infrastructure scaling required.
- Architected a full REST API surface (**15+ endpoints**) with **JWT + RBAC** from zero to production, structured so new features plug in without touching existing logic.

EDUCATION

Vellore Institute of Technology, Chennai Campus, TN Sep 2022 – Present

B.Tech Computer Science — Specialization in AI & Robotics

CGPA: 7.64

ADDITIONAL INFORMATION

Certifications: Microsoft Azure Data Fundamentals (DP-900) & MongoDB Certified Developer

Activities: *International Committee Member* — Led logistics for 4+ global events; built working relationships with international clients across cultures.